0136. Single Number

Easy | Category: Bit Manipulation | LeetCode: https://leetcode.com/problems/sing

Problem

- Every element appears twice except one. Find it in O(n) time and O(1) space.

Intuition

- XOR is its own inverse (a ^ a = 0) and order does not matter, so XOR-ing everything cancels the pairs and leaves the single value.

Brute Force

- HashSet add/remove: O(n) space.

Optimal Approach

- X = 0; for each v: x ^= v. Return x.

Worked Example

```
[4,1,2,1,2].

0^4=4  4^1=5  5^2=7  7^1=6  6^2=4  -> 4
```

Java Solution

```java
public class SingleNumber {
    public int singleNumber(int[] nums) {
        int x = 0;
        for (int v : nums) x ^= v;
        return x;
    }
}
```

Complexity

- Time: O(n). Space: O(1).

Pitfalls

- Works for negatives too, because XOR is bitwise.